# Fluent

PENG

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## 1 Introduction

Fluent User's Guide

**Definition 1.1** (Ansys Fluent)**.** *Ansys Fluent a computer program written in C for modelling fluid flow, heat transfer, and chemical reactions in complex geometries.*

## 2 Basic Steps for CFD Analysis

1. define the modeling goals: simplifying assumptions, boundary conditions
2. create the model geometry and mesh:
   - unstructured meshes, default units: meters
   - mesh resolution sufficient in each region of the domain
   - mesh check: cell skewness $< 0.98$
3. set up the solver and physical models
   - boundary conditions at all boundary zones
   - initial solution
   - converged solution is not necessarily a correct one
   - use **second-order upwind discretization** for better accuracy
4. compute and monitor the solution
   - number of iterations to reach a converged solution
5. examine and save the results
   - flow features: flow pattern, separation, .etc
   - quantitative results: forces and moments
6. revisions to the numerical or physical model parameters

## 3 Executing Ansys Fluent